

E-Commerce Sales & Customer Analytics

SQL-Based Business Analysis Report

Brazilian E-Commerce Public Dataset (Olist)

This report summarizes the business insights obtained through exploratory SQL, advanced SQL, customer analytics, product and category analysis, seller and regional analysis, time-series analysis, and business-question-driven investigation.

1. Executive Summary

The analysis covers 99,441 orders and 99,441 customer records from the Olist Brazilian e-commerce dataset. The platform demonstrates strong fulfillment performance, but customer retention is the clearest strategic weakness.

Delivered orders generated R\$13.22M in merchandise revenue. The delivery rate was 97.02%, indicating a strong operational foundation.

Only about 3.00% of customers were repeat customers. Repeat customers, however, had a substantially higher average value (R\$260.05) than one-time customers (R\$137.96).

Revenue is concentrated across customers and product categories, while cross-state commerce represents a major share of revenue.

- Prioritize retention and repeat-purchase programs.
- Protect high-value customers and use targeted re-engagement campaigns.
- Strengthen the highest-contributing product categories while testing growth opportunities.
- Use regional and seller-level analytics to improve marketplace coverage and efficiency.

2. Overall Business Performance

The dataset contains 99,441 orders. Of these, 96,478 were delivered, producing a delivery rate of 97.02%.

Delivered merchandise revenue totaled R\$13,221,498.11. This figure represents item prices and excludes freight from merchandise revenue.

The analysis uses delivered orders as the primary basis for revenue-focused business analysis so that cancelled or otherwise incomplete transactions do not distort performance measures.

- Total orders: 99,441
- Delivered orders: 96,478
- Delivery rate: 97.02%
- Delivered merchandise revenue: R\$13.22M

3. Customer Insights

Customer behavior shows a strong one-time-purchase pattern. Repeat customers account for only 3.00% of customers, while the large majority purchase only once.

Despite their small number, repeat customers are considerably more valuable. Their average customer value is R\$260.05 compared with R\$137.96 for one-time customers.

One-time customers contribute approximately 94.49% of revenue, while repeat customers contribute 5.51%. This indicates a large opportunity to increase lifetime value through retention.

Revenue concentration is also significant: the top 20% of customers generate approximately 56.62% of total revenue. Approximately 42,009 customers are needed to reach 80% of cumulative revenue.

- Repeat customer rate: 3.00%
- Repeat customer average value: R\$260.05
- One-time customer average value: R\$137.96
- Top 20% customer revenue share: 56.62%
- Customers needed for ~80% cumulative revenue: 42,009

4. Product & Category Insights

Revenue is distributed across many product categories, but a relatively small group of categories drives a large share of business.

The five leading categories account for approximately 39.83% of revenue, while the top ten categories account for approximately 62.43%.

The leading categories are health_beauty (9.33%), watches_gifts (8.82%), bed_bath_table (7.74%), sports_leisure (7.22%), and computers_accessories (6.72%).

Category-level growth analysis was also used to distinguish meaningful growth from noise by focusing on categories with at least R\$50,000 of 2017 revenue.

- Top 5 category revenue share: 39.83%
- Top 10 category revenue share: 62.43%
- Largest category: health_beauty — 9.33%
- Second: watches_gifts — 8.82%
- Third: bed_bath_table — 7.74%

5. Seller & Regional Insights

Seller analysis shows that the marketplace is not excessively dependent on a few individual sellers. The top seller contributes about 1.72% of revenue, while the top five and top ten sellers contribute 7.71% and 13.27%, respectively.

Regional concentration is more pronounced. São Paulo is the leading customer market with approximately R\$5.07M in customer-side revenue, while São Paulo sellers generate approximately R\$8.51M.

Cross-state commerce accounts for approximately 68.73% of revenue, demonstrating that marketplace performance depends heavily on transactions between different Brazilian states.

- Top seller revenue share: 1.72%
- Top 5 seller revenue share: 7.71%
- Top 10 seller revenue share: 13.27%

- Cross-state revenue share: 68.73%
- Leading seller state by revenue: São Paulo

6. Time-Series & Growth Insights

Time-series SQL was used to analyze monthly revenue, order volume, month-over-month growth, running revenue totals, average order value, yearly performance, revenue rankings, and year-over-year growth.

Between 2017 and 2018, revenue increased by approximately 21.05% and order volume increased by roughly 21.5%. Average order value remained broadly stable and slightly declined.

The combination of higher order volume with relatively stable average order value suggests that growth was driven primarily by increased transaction volume rather than a major increase in basket size.

- 2018 revenue growth: approximately 21.05%
- 2018 order growth: approximately 21.5%
- AOV: broadly stable/slightly lower

7. Key Business Problems

The largest issue is low customer retention. A repeat rate of only 3.00% means most customers do not return for another purchase.

Revenue concentration creates customer-level risk because a relatively small portion of customers contributes a disproportionately large share of revenue.

Category concentration means that a limited set of product groups accounts for much of marketplace revenue, making category performance important to overall results.

Cross-state commerce is a major revenue driver, increasing the importance of regional seller coverage, logistics, and delivery consistency.

8. Data-Driven Recommendations

1. Build a retention program: segment customers after their first purchase and trigger personalized offers, reminders, and category recommendations.
2. Protect high-value customers: identify high-spend and repeat customers and provide targeted benefits rather than blanket discounts.
3. Increase cross-sell opportunities: recommend complementary products based on previous category purchases and customer value.
4. Strengthen leading categories: maintain availability and seller coverage in the highest-revenue categories while investing selectively in growing categories.
5. Improve regional strategy: use customer-state and seller-state combinations to identify strong routes and underserved markets.
6. Monitor seller concentration and efficiency: track revenue per order and seller contribution to maintain marketplace resilience.
7. Continue time-series monitoring: track monthly revenue, order growth, AOV, and category trends to detect changes early.

9. Final Analyst Conclusion

The SQL analysis indicates that the marketplace has a strong operational base and demonstrated meaningful growth, but its biggest opportunity is customer retention.

The business can potentially increase customer lifetime value without relying solely on acquiring new customers by converting one-time buyers into repeat customers.

The analysis also shows that customer value, category contribution, regional flows, seller efficiency, and time-series behavior can be combined into a practical analytics framework for ongoing business decision-making.

Overall, the recommended strategy is to protect high-value demand, improve retention, strengthen core categories, and use regional and seller-level data to guide marketplace expansion.

Analysis scope: SQL-based analysis of the Olist Brazilian E-Commerce Public Dataset. Revenue figures refer to merchandise price unless otherwise stated.